

The Novel DeepBoost Multi Output Regressor for Household Finance Intelligence

S. H. S. V. Suwandarathna

Department of Computer Science and Engineering
Informatics Institute of Technology
No 10, Trelawney PI, Colombo 00400, Sri Lanka
savinthie.20210181@iit.ac.lk

G. O. Priyankara

Department of Computer Science and Engineering
Informatics Institute of Technology
No 10, Trelawney PI, Colombo 00400, Sri Lanka
obhasha@live.com

Abstract—This paper forecasts the microeconomic level household expenditures using a novel hybrid deep learning approach. In terms of research significance, household finance control has a major influence on the finance system within the economy. Accurate forecasting of household finances assists in maintaining positive financial behavior among individuals and the economy. The DeepBoost multi-output regressor proposed in this paper is based on the 1D CNN-ANN and the XGBoost. The proposed model in this paper is compared with the R^2 , MSE, and MAE since it's a regression problem. The experimental results reveal that the proposed DeepBoost multi-output regressor has the best application in forecasting the multiple expenditures of households by outperforming the ANN, 1D CNN-ANN, and Random Forest Regressor models. The proposed DeepBoost multi-output regressor evaluated the housing, food, transportation, healthcare, other necessities, childcare, and tax expenditures that had 0.94, 0.98, 0.83, 0.94, 0.97, 0.97, and 0.99 values for the R^2 , 9037.71, 2692.12, 9788, 15077.33, 1373.93, 13629.36, and 1904.52 values for the MSE, and 66.07, 34.05, 73.17, 87.05, 26.25, 78.74, and 29.47 MAE values than the ANN, RFR, and 1D CNN-ANN models.

Keywords - DeepBoost, Household finance, Hybrid deep learning model, Multi-output regressor, XGBoost

I. INTRODUCTION

The household finances triggered the stability of the financial system through its financial health and the behavior patterns of credit on households [1]. A very small number of Machine Learning (ML) and Deep Learning (DL) technologies are utilized in household finances in the existing studies. ML is simply used to solve problems using trained data while DL utilizes neural networks to learn from large and complex datasets to solve problems [2], [3], [4]. This paper aimed to implement the proposed DeepBoost multi-output regressor, a novel hybrid DL model to forecast the multiple expenditures of the household. The proposed DeepBoost multi-output regressor is a combination of the 1D CNN-Artificial Neural Network (ANN) hybrid DL model and an XGBoost model. This paper will provide a piece of in-depth information on the proposed DeepBoost multi-output regressor. The rest of the paper is mainly divided into Aims and objectives (Section II), Literature survey (Section III), Methodology (Section IV),

Results and discussion (Section V), Limitations and future work (Section VI), and Conclusion (Section VII).

II. AIMS AND OBJECTIVES

A. Aims

This research primarily aims to implement the proposed DeepBoost multi-output regressor by combining the hybrid 1D CNN-ANN model with the XGBoost model. The 1D CNN-ANN model provides a new aspect to deep learning technologies due to its hybrid technique. The primary contribution of the research is to introduce a novel hybrid DL technique to household finance forecasting.

B. Objectives

The objectives of this research is as follows:

- To develop the hybrid DL model by combining a 1D CNN and an ANN model.
- To develop the DeepBoost multi-output regressor by integrating the hybrid DL model with the XGBoost model.
- To evaluate the performance of the DeepBoost regressor model with a Random Forest regressor, ANN, and the 1D CNN-ANN models.
- To predict the housing, food, transportation, healthcare, other necessities, childcare, and tax expenditures using the DeepBoost regressor.

III. LITERATURE SURVEY

A. Machine Learning for Household studies

In recent years, ML models have been utilized for multiple applications in households for the regression and classification tasks.

In 2020, an Ordinary Least Square (OLS) regression model was employed to analyze the impact of fiscal expenditure on the consumption expenditure of rural and urban households [5].

TABLE I. RESULTS OF THE EXISTING RFR

Model	Accuracy
Decision Tree	48%
Random Forest Regressor	74.35%

In 2021, a Random Forest Regression (RFR) algorithm was employed to forecast the annual income of the household by the provided household expenditures by outperforming a Decision Tree (DT). TABLE I depicts the prediction accuracy of the two models [6].

In 2022, several ML algorithms were proposed for household research. An existing study utilized five OLS regression models to investigate the degree to which low and moderate-income household individuals experience consistent feelings of financial stability over six months [7]. A proposed Extended Linear Expenditure System (ELES) was employed to identify the regression outcomes for household expenditures such as food and cooking wine, clothing, living, daily necessities and services, transportation and communication, education, culture and entertainment, healthcare, and other supplies and services [8]. Least Square Regression (LSR) and Ridge Regression (RR) models were utilized to analyze the impact of the national economy, price index, and household income level on the consumption of households [9]. An improved version of the Light Gradient Boosting Machine (LightGBM) model was utilized to investigate the poverty status of the households. The LightGBM included filter, wrapper, embedded, and hybrid techniques as its classification category in the feature selection stage. Before the finalization of the LightGBM model, an ANN model was implemented and compared. TABLE II depicts the reported results at the evaluation stage proving that improved version of the the LightGBM outperformed the ANN model in the existing study [10].

TABLE II. RESULTS OF THE EXISTING IMPROVED LIGHTGBM

Model	F1-score	Recall	Precision	Log loss
ANN	0.94	0.76	0.80	0.146
LightGBM	0.95	0.83	0.82	0.137

ML algorithms such as Gradient Boosted Tree (GBT), Multi-Layer Perceptron Neural Network (MLPNN), Naïve Bayes, RFR, and Decision Tree (DT) were used as proposed models to predict the financially calculative households to identify the households with a lower income level and finally proved that the GBT outperforms the other proposed models [11]. A proposed Focused Embedded Logistic Regression (FeLR) model was employed to investigate and solve the variance and rarity of the likely poor households [12]. The XGBoost, LightGBM, and CatBoost algorithms were proposed to recognize poverty within the households where the XGBoost model outperformed the other two ensemble learning models after conducting 1500 iterations as depicted in TABLE III [13].

TABLE III. RESULTS OF THE EXISTING XGBOOST

Model	Accuracy	Recall	F1-score
CatBoost	95.8%	91.2%	0.89
LightGBM	96.1%	95.5%	0.90
XGBoost	98.2%	95.2%	0.95

In 2024, the proposed Multiple regression and Robust regression techniques examined the subjective financial suc-

cess of the households [14]. In conclusion, the overall ML techniques associated with households up to date are considered traditional ML techniques that contributes to household finances.

B. Deep Learning for Household studies

In recent years, DL models have rarely been used for applications in households for regression and classification tasks.

In 2021, a proposed ANN model was utilized to predict the savings-to-income and debt-to-asset household financial ratios. The ANN model was compared with an OLS regression model and proved to be more robust with the additional use of Stata 15.1 software by minimizing the error variance in the prediction [15]. In conclusion, it is justified that the existing research did not utilize any hybrid DL techniques to contribute to household finance.

IV. METHODOLOGY

A. Data Collection

This stage requires the collection of household data from the open sources. The secondary data collection methodology was utilized in this research due to the lack of available household datasets. The utilized secondary household dataset consisted of 31430 household records relevant to the year 2023 that were collected from the currently existing open source ‘Economic Policy Institute Family Budget Calculator, USA’. The monetary representation of the data in the available secondary household dataset is represented using dollars [16].

B. Data Preprocessing

This stage requires cleaning the dataset by handling the missing values and inconsistent data. Most importantly, the dataset was split into three categories as training, testing, and validation sets at the cross validation stage, which is discussed in the Implementation of the DeepBoost Multi Output Regressor (Subsection D).

C. Feature Engineering

This stage requires the engineering of features by selecting the most important features and deriving new features from the existing data. ‘Total expenditure’, ‘Median family income’, ‘State abbreviation’, ‘County’, ‘Number of counties in the state’, ‘Family’, ‘Food’, ‘Transportation’, ‘Housing’, ‘Healthcare’, ‘Taxes’, ‘Childcare’, and ‘Other necessities’ are the most important features and the output labels available within the dataset. Derived features such as ‘Number of parents’, ‘Number of children’, and ‘Number of family members’ were created with the use of the ‘Family’ feature. Additionally derived features and output labels such as ‘Income sufficiency indicator’, ‘Cost per member’, ‘Cost per child’, ‘Cost per parent’, and ‘Other cost’ were created with the use of the ‘Median family income’, ‘Total expenditure’, and ‘Number of family members’. Furthermore, the final predictions of the DeepBoost regressor model included the ‘Food’, ‘Transportation’, ‘Housing’, ‘Healthcare’, ‘Taxes’, ‘Childcare’, and

‘Other necessities’ as multiple outputs. TABLE IV depicts the summary of the features engineered. At the end of this stage, the feature scaling technique was utilized as a preprocessing step.

TABLE IV. FEATURE AND OUTPUT LABEL DESCRIPTION

Feature and output label Name	Existing	Derived
Total expenditure	Yes	-
Median family income	Yes	-
State abbreviation	Yes	-
County	Yes	-
Number of counties in the state	Yes	-
Family	Yes	-
Food	Yes	-
Housing	Yes	-
Healthcare	Yes	-
Taxes	Yes	-
Childcare	Yes	-
Other necessities	Yes	-
Number of parents	-	Yes
Number of children	-	Yes
Number of family members	-	Yes
Income sufficiency indicator	-	Yes
Cost per member	-	Yes
Cost per child	-	Yes
Cost per parent	-	Yes
Other cost	-	Yes

D. Implementation of the DeepBoost Multi Output Regressor

The DeepBoost architecture is implemented by combining the hybrid 1D CNN-ANN model and an XGBoost ensemble learning model. An ANN is a simple Feed Forward Neural Network (FFNN) that turns down the error between the predicted and the expected output by carrying out backpropagation to fine-tune the weights and biases in the Neural Network (NN) and is very effective in regression and classification-related tasks [17], [18], [19]. The 1D CNN has been proven to be specialized in capturing localized data patterns within the tabular dataset, which has won recent interest in the finance domain [20]. The approach of the input layer of the hybrid DL model is to accept the well-defined and reshaped features. The approach of the 1D CNN module is to capture the localized data patterns within the input features of the tabular household dataset using the Conv1D layer with the ‘ReLU’ activation function followed by the ‘Flatten’ layer. The ANN module processes the features extracted from the ‘Flatten’ layer through the two dense layers with 128 and 64 neurons along with the ‘ReLU’ activation function, Batch Normalization, and Dropout regularization, enabling the model to recognize and learn the complex and nonlinear relationships while avoiding overfitting [21], [22], [23], [24]. The output layer of the hybrid DL model was implemented independently with the ‘Linear’ activation function to produce each label separately. The cross-validation data sampling technique was used to produce a generalized model [25]. The K-fold cross-validation technique was applied on the train, test, and validation data representing the 80:10:10 ratio to reduce the bias and the variance for a robust hybrid model evaluation with 5 folds. Additionally, the

‘Adam’ optimizer was used for the optimization of the neural networks within the hybrid DL model [26]. The hybrid DL model was implemented using a multi-output regression technique, which enabled the hybrid DL model to predict multiple outputs [27]. Since, XGBoost is a tree ensemble algorithm that is majorly compatible with tabular data problems and are used to generate new models from the previous models’ residuals to make the final prediction by merging the models, the DeepBoost Multi Output Regressor architecture integrated the developed 1D CNN-ANN DL model with the XGBoost model to provide a better performing model [28], [29]. Fig. 1. illustrates high-level architecture of the DeepBoost multi-output regressor model and TABLE V represents the multiple outputs generated as the final prediction. The generalized version of the DeepBoost multi-output regressor is illustrated in Fig. 2. to Fig. 6. depicting the training and validation loss curve for the 5 folds proving that the model is not overfitting or underfitting.

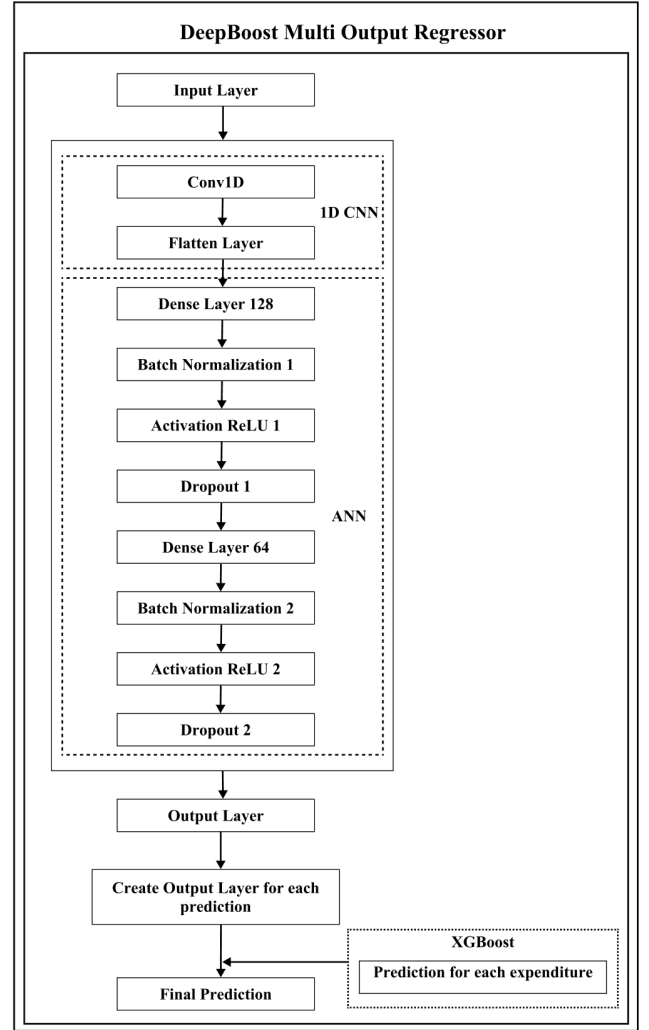


Fig. 1. High-level architecture of the DeepBoost multi-output regressor combining the 1D CNN-ANN hybrid model and the XGBoost model.

TABLE V. MULTIPLE OUTPUTS OF THE FINAL PREDICTION

Predicted output
Housing
Food
Transportation
Healthcare
Other necessities
Childcare
Taxes

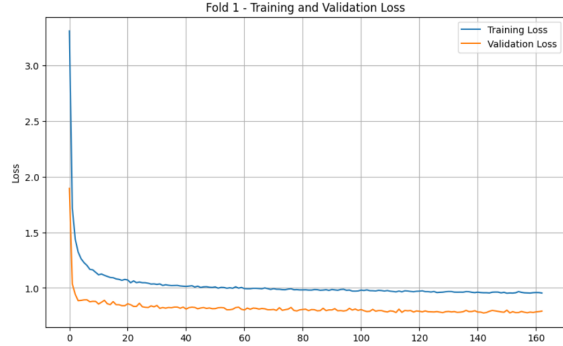


Fig. 2. Training and validation of fold 1 in DeepBoost.

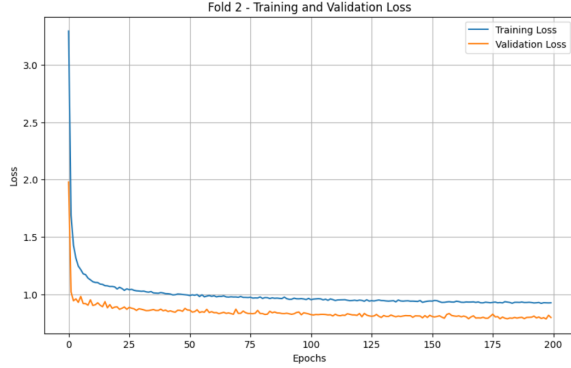


Fig. 3. Training and validation of fold 2 in DeepBoost.

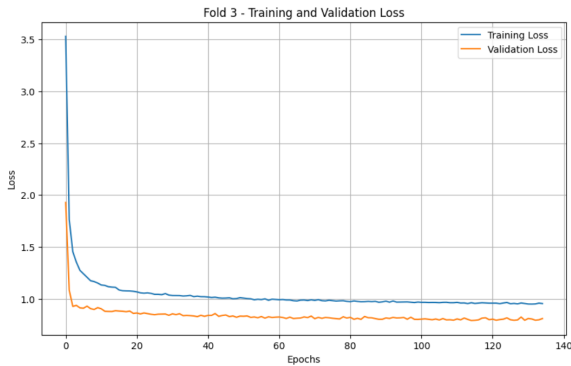


Fig. 4. Training and validation of fold 3 in DeepBoost.

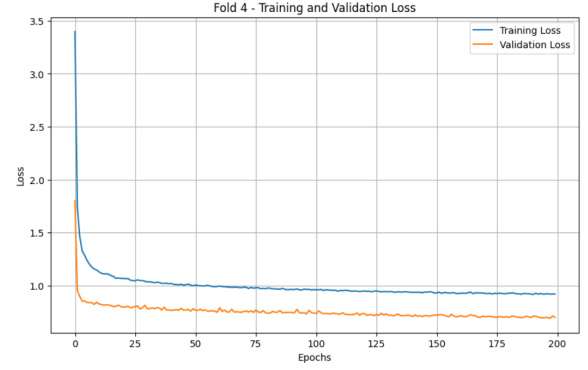


Fig. 5. Training and validation of fold 4 in DeepBoost.

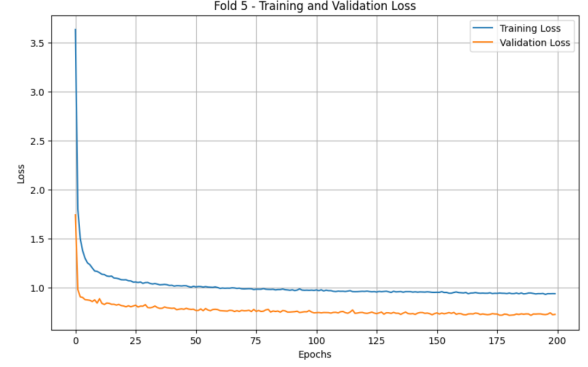


Fig. 6. Training and validation of fold 5 in DeepBoost.

E. Evaluation

The standard evaluation metrics that were used to identify the performance of the proposed models were the R^2 , MSE, and MAE [30], [31], [32]. A brief description of each metric is provided as follows:

- R^2 is used to measure the model's goodness of fit.
- MSE, also known as Mean Squared Error, is used to measure the mean square error between the actual and the predicted values.
- MAE, also known as Mean Absolute Error, is used to measure the difference in the mean absolute error between the actual and predicted values.

V. RESULTS AND DISCUSSION

The performance results of the ANN, RFR, 1D CNN-ANN, and the DeepBoost models are represented in TABLE VI to TABLE XII.

TABLE VI. TEST RESULTS OF HOUSING

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.8820	0.8852	0.9178	0.94
MSE	18017.49	17529.03	12550.45	9037.71
MAE	93.70	91.69	74.10	66.07

TABLE VI depicts that the DeepBoost had 0.94, 9037.71, and 66.07 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models where it outperformed the R^2 , MSE, and MAE values than the ANN, 1D CNN-ANN, and RFR for Housing.

TABLE VII. TEST RESULTS OF FOOD

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.9584	0.9590	0.9692	0.98
MSE	4812.27	4739.41	3555.12	2692.12
MAE	46.64	46.24	38.79	34.05

TABLE VII depicts that the DeepBoost had 0.98, 2692.12, and 34.05 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models where it outperformed the R^2 , MSE, and MAE values than the ANN, 1D CNN-ANN, and RFR for Food.

TABLE VIII. TEST RESULTS OF TRANSPORTATION

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.6839	0.6844	0.7446	0.83
MSE	17812.88	17784.09	14391.54	9788
MAE	98.59	98.47	86.69	73.17

TABLE VIII depicts that the DeepBoost had 0.83, 9788, and 73.17 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models where it outperformed the R^2 , MSE, and MAE values than the ANN, 1D CNN-ANN, and RFR for Transportation.

TABLE IX. TEST RESULTS OF HEALTHCARE

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.8259	0.8343	0.9313	0.94
MSE	41512.44	39491.13	16392.38	15077.33
MAE	146.78	141.60	84.24	87.05

TABLE IX depicts that the DeepBoost had 0.94, 15077.33, and 87.05 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models where it outperformed the R^2 , MSE, and MAE values than the ANN and 1D CNN-ANN. Furthermore, it was noted that the RFR outperformed the MAE value than the DeepBoost for Healthcare.

TABLE X. TEST RESULTS OF OTHER NECESSITIES

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.9463	0.9472	0.9632	0.97
MSE	28822.22	2771.97	1933.98	1373.93
MAE	37.67	37.09	30.29	26.25

TABLE X depicts that the DeepBoost had 0.97, 1373.93, and 26.25 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models where it outperformed the R^2 , MSE, and MAE values than the ANN, 1D CNN-ANN, and RFR for Other necessities.

TABLE XI depicts that the DeepBoost had 0.97, 13629.36, and 78.74 values for the R^2 , MSE, and MAE compared to

the ANN, 1D CNN-ANN, and RFR models. Furthermore, it was noted that the RFR outperformed the MAE value than the DeepBoost for Childcare.

TABLE XI. TEST RESULTS OF CHILDCARE

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.9186	0.9224	0.9636	0.97
MSE	32160.61	30618.51	14361.07	13629.36
MAE	124.24	122.40	70.99	78.74

TABLE XII depicts that the DeepBoost had 0.99, 1904.52, and 29.47 values for the R^2 , MSE, and MAE compared to the ANN, 1D CNN-ANN, and RFR models. Furthermore, it was noted that the RFR outperformed the MAE value than the DeepBoost for Taxes.

TABLE XII. TEST RESULTS OF TAXES

Evaluation Metrics	ANN	1D CNN-ANN	RFR	DeepBoost
R^2	0.9578	0.9595	0.9876	0.99
MSE	7172.98	6864.87	2081.79	1904.52
MAE	58.69	58.47	21.83	29.47

Therefore, the proposed DeepBoost multi-output regressor model is considered the best model overall.

VI. LIMITATIONS AND FUTURE WORK

While the proposed DeepBoost multi-output regressor demonstrates significant effective model performance, it has several limitations. As discussed in the Methodology (Section IV), the primary limitation of this study is the lack of household data availability. It is suggested to collect household data that can create a large household dataset or to implement an open-source household dataset that will be useful to conduct future research on the interest of ML and DL on household finance control. The secondary limitation of this study is the lack of time series household data. If time series household data is available, the current research interest can be further benchmarked with a few other DL algorithms, such as Recurrent Neural Network (RNNs) and Long Short Term Memory (LSTMs), as they are well-supported with time series data. Since the proposed DeepBoost multi-output regressor is a DL model, it is suggested to include Explainable AI (XAI) for better black box model interpretability for future work as it will be beneficial for stakeholders of the Financial and DL technologies [33], [34].

VII. CONCLUSION

The proposed DeepBoost multi-output regressor is a novel approach introduced to household studies that is a combination of the 1D CNN-ANN and an ensembled architecture to predict multiple outputs that are well-suited for regression problems. The key idea within the DeepBoost multi-output regressor is that it is a novel hybrid DL model that is used to predict multiple household expenditures for the given inputs by addressing the lack or no usage of hybrid DL models within household finance studies.

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